Q.1. What are logic gates?

Ans. Logic gates are the fundamental building blocks of digital systems. They are circuits in which the output depends upon the inputs according to some logic rules.

Q.2. What is a logic circuit?

Ans. A logic circuit is an electronic circuit that operates on digital signals in accordance with a logic function.

Q.3. What is logic design?

Ans. The interconnection of gates to perform a variety of logical operations is called logic design.

Q.4. What is a truth table?

Ans. A truth table is a table which lists all possible combinations of inputs and the corresponding outputs.

Q.5. What is a positive logic system?

Ans. A positive logic system is one in which the higher of the two voltage levels represents logic 1 and the lower of the two voltage levels represents logic 0.

Q.6. What is a negative logic system?

Ans. A negative logic system is one in which the higher of the two voltage levels represents logic 0 and the lower of the two voltage levels represent logic 1.

Q.7. What is the maximum number of outputs a logic gate can have?

Ans. All logic gates can have only one output.

Q.8. What do you mean by level logic?

Ans. Level logic means voltage levels represent logic 0 and logic 1.

Q.9. What is an AND gate?

Ans. An AND gate is a logic gate with two or more inputs and only one output. It outputs a 1 only when each one of its in is 1. It outputs a 0 even if one of its inputs is a 0. It is a basic gate. It is also called an all or nothing gate. It performs AND operation.

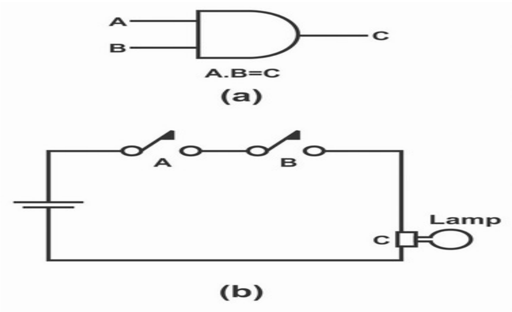


Fig 2.11

Q.10. Which gate is called an all or nothing gate? Why?

Ans. An AND gate is called an all or nothing gate; because it produces a 1 only in one case when all its inputs are a 1. In all other cases its output is a zero.

Q.11. What is an OR-gate?

Ans. An OR gate is a logic gate with two or more inputs and only one output. It outputs a 1 even if one of its inputs is a 1. It outputs a 0 only when each one of its input is a 0. It is also called an any or all gate. It is a basic gate. It performs OR operation.

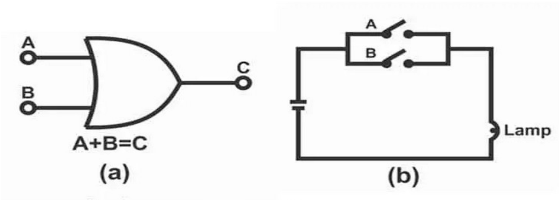


Fig 2.4

Q.12. Which gate is called any or all gate? Why?

Ans. An OR gate is called any or all gate; because it produces a 1 even if one of its inputs is a 1. It produces a 0 only when all the inputs are a 0.

Q.13. What is a NOT gate?

Ans. A NOT gate is a logic gate with only one input and one output. Its output is always the complement of its input. It is a basic gate. It performs NOT operation.

Q.14. What is an inverter?

Ans. An inverter is nothing but a NOT gate.

Q.15. Which gates can be used as inverters in addition to the NOT gate?

Ans. NAND gate, NOR gate. X-OR gate and X-NOR gate can also be used as inverters in addition to the NOT gate.

Q.16. Why AND, OR and NOT gates are called basic gates?

Ans. AND, OR, and NOT gates are called basic gates or basic building blocks because any digital circuit of any complexity can be built by using only these three gates.

17. Explain what is a combinational circuit?

In a combinational circuit, the output depends upon present input(s) only i.e, not dependant on the previous input(s). The combinational circuit has no memory element. It consists of logic gates only.

18. Write two characteristics of combinational circuits.

The two characteristics of combinational circuits are:

In combinational circuits, the output exists as long as the input exists.

A combinational circuit will always respond in the same fashion to the input function, when we apply signal to the input terminal of the combinational logic circuit.

19. Explain what is a half-adder?

A logic circuit, that can add two 1-bit numbers and produce outputs for sum and carry, is called a half-adder.

20. Explain what is a full-adder?

A binary adder, which can add two 1-bit binary numbers along with a carry bit and produces outputs for sum and carry is called a full-adder.

21. Explain what is a flip-flop?

A flip-flop is a basic memory element that is made of an assembly of logic gates and is used to store 1-bit of information.

22. Explain what is a latch?

It is a D-type of flip-flop and stores one bit of data.

23. Explain what is an excitation table?

Excitation table gives an information about Explain what should be the flip-flop inputs if the outputs are specified before and after the clock pulses.

24. Explain what is a state table?

State table consists of complete information about present state, next state, and outputs of a sequential circuit.

25. Explain what is Boolean Algebra?

Boolean algebra is a mathematics system of logic in which truth functions are expresses as symbols and then these symbols are manipulated to arrive at conclusion.

26. Explain what are the basic logic elements?

Basic logic elements are NOT gate, AND gate, OR gate and the flip-flop.

27. Explain what is a truth table?

Truth table is a table that gives outputs for all possible combinations of inputs to a logic circuit.

28. Define positive logic and negative logic.

If the higher of the two voltages represents a 1 and the lower voltage represents a 0, then the logic is called a positive logic. On the other hand, if the lower voltage represents a 1 and the higher voltage a 0, we have a negative logic.

29. Explain what is pulse logic system?

A logic system in which a bit is recognized by the presence or absence of a pulse is called a pulse or dynamic logic system.

30. Explain what is an inverter?

An inverter is a logic gate whose output is the inverse or complement of its input.

31. Explain what are the universal logic gates?

Universal gate is a gate that can perform all the basic logical operations such as NAND and NOR gates.

32. Explain what is the specialty of NAND and NOR gates?

The specialty of NAND and NOR gates is that they are universal gates and can perform all the basic logical operations.

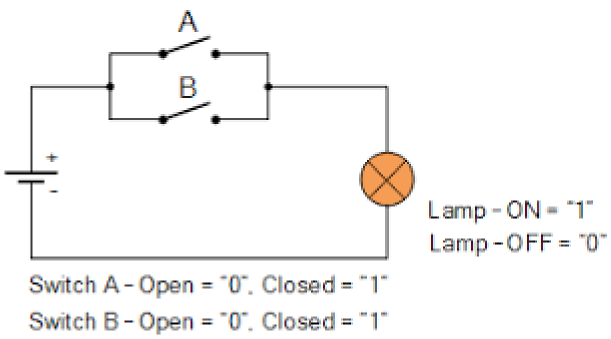
34. Explain why NAND-NAND realization is preferred over AND-OR realization?

NAND-NAND realization needs only one type of gate(NAND), that minimizes IC package counter.

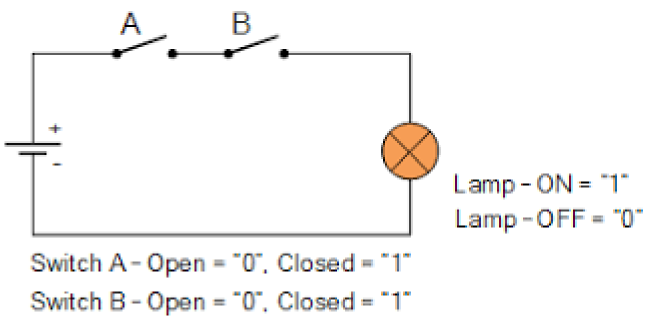
35. Explain why is a two-input NAND gate called universal gate?

NAND gate is called universal gate because any digital system can be implemented with the NAND gate. Sequential and combinational circuits can be constructed with these gates because element circuits like flip-flop can be constructed from two NAND gates connected back-to-back. NAND gates are common in hardware because they are easily available in the ICs form. A NAND gate is in fact a NOT-AND gate. It can be obtained by connecting a NOT gate in the output of an AND gate.

36.OR gate as a switch:



37.AND gate as a switch:



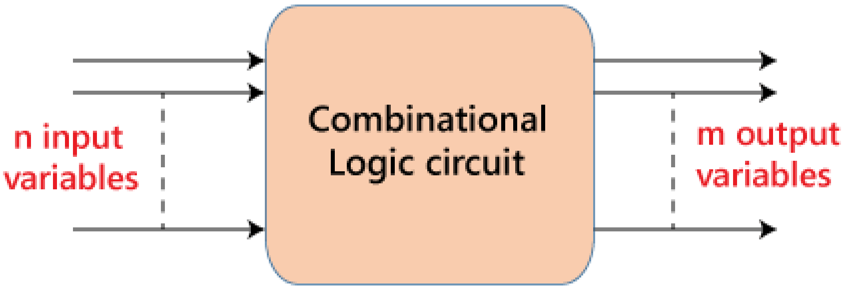
Questions On Combinational logic Circuits

Combinational Logic circuits

The combinational logic circuits are the circuits that contain different types of logic gates. Simply, a circuit in which different types of logic gates are combined is known as a **combinational logic circuit**. The output of the combinational circuit is determined from the present combination of inputs, regardless of the previous input. The input variables, logic gates, and output variables are the basic components of the combinational logic circuit. There are different types of combinational logic circuits, such as Adder, Subtractor, Decoder, Encoder, Multiplexer, and De-multiplexer.

There are the following characteristics of the combinational logic circuit:

- At any instant of time, the output of the combinational circuits depends only on the present input terminals.
- The combinational circuit doesn't have any backup or previous memory. The present state of the circuit is not affected by the previous state of the input.
- The n number of inputs and m number of outputs are possible in combinational logic circuits.



Block diagram of a combinational circuit

The 'n' input variable comes from the external source while the 'm' output variable goes to the external destination. In many applications, the source or destinations are storage registers.

Half Adder

The half adder is a basic building block having two inputs and two outputs. The adder is used to perform OR operation of two single bit binary numbers. The **carry** and **sum** are two output states of the half adder.

Full Adder

The half adder is used to add only two numbers. To overcome this problem, the full adder was developed. The full adder is used to add three 1-bit binary numbers A, B, and carry C. The full adder has three input states and two output states i.e., sum and carry.

Half Subtractors

The half subtractor is also a building block of subtracting two binary numbers. It has two inputs and two outputs. This circuit is used to subtract two single bit binary numbers A and B. The **'diff'** and **'borrow'** are the two output state of the half adder.

AD

Full Subtractors

The Half Subtractor is used to subtract only two numbers. To overcome this problem, full subtractor was designed. The full subtractor is used to subtract three 1-bit numbers A, B, and C, which are **minuend, subtrahend**, and **borrow,** respectively. The full subtractor has three input states and two output states i.e., diff and borrow.

Multiplexers

The multiplexer is a combinational circuit that has n-data inputs and a single output. It is also known as the **data selector** which selects one input from the inputs and routes it to the output. With the help of the selected inputs, one input line from the n-input lines is selected. The enable input is denoted by E, which is used in cascade.

De-multiplexers

A De-multiplexer performs the reverse operation of a multiplexer. The de-multiplexer has only one input, which is distributed over several outputs. One output line is selected at a time by selecting lines. The input is transmitted to the selected output line.

Decoder

A decoder is a combinational circuit having n inputs and to a maximum of m = 2n outputs. The decoder is the same as the de-multiplexer. The only difference between de-multiplexer and decoder is that in the decoder, there is no data input. The decoder performs an operation that is completely opposite of an encoder.

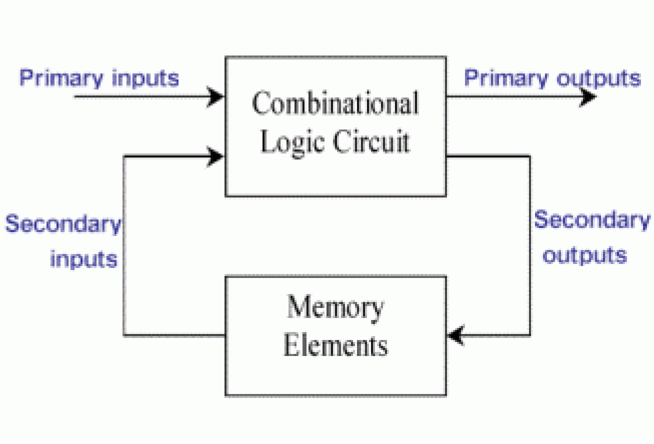
Encoder

The encoder is used to perform the reverse operation of the decoder. An encoder having n number of inputs and m number of outputs is used to produce m-bit binary code which is related to the digital input number. The encoder takes the digital word and converts it into another digital word.

Questions On Sequential Circuits

Q.1. What is sequential circuit?

Ans. These are defined as digital circuit whose output is dependent not only on the present input value but also on the past history of its input. The sequential Circuits are designed using the combinational circuits along with a memory devices known as Flip-Flops. The sequential Circuits depend over the input value as well as the stored levels.



Sequential Circuits block diagram

Q.2. What is a flip flop?

Ans. Flip flop is one bit storage bistable device. Flip flop is also called latch. It stores binary value. It is the basic building block of the digital electronic systems. These are the basically the data storing devices which store the information of two stable states of the system. A flip-flop stores only a single bit of data at a time.

Q.3. What is the difference between latch and flip flop?

Ans. The difference between the latch and flip flop is the method of changing their state. The latch do not have clock input but flip flop has a clock input.

Parameter	Latch	Flip Flop
AREA	LESS	MORE
POWER	LESS	MORE
DESIGN ROBUSTNESS	DESIGN IS NOISY	DESIGN IS ROBUST

Latch Vs Flip Flop

Q.4. Does sequential circuit contain memory element?

Ans. Yes, sequential circuit contain memory element. A storing element is added to store the various stable state levels information. This help in relating the feedback data from the past to the present data.

Q.5. What is the application of T flip flop?

Ans. T Flip Flops can be used as follows-

(a) Frequency divider
(b) Counters
(c) Binary Addition devices

Q.6. How race around condition can be eliminated?

Ans. It is essential to understand the race around condition before the development of edge triggered flip flop. As we know that the conditions s==1 and R==1 are not allowed in flip flop by the use of feedback correction. Under this situation when input J and K are 1 and 1 output will change from 0 to 1. To avoid race around condition,we use master slave flip flop. It has two different flip flop, which are connected serially.

Q.7. What is a counter?

Ans. Counter is a sequential circuit which is used to count the number of clock pulses of the circuit. It is also sometimes used to display the number of time any event is repeated or happening. It's a simple counter in the digital logics and computation which calculates the number of times an assigned event is taking place.

Q.8. What is the difference between asynchronous counter and synchronous counter ?

Ans. Asynchronous counter's speed is less while synchronous counter's speed is high. In asynchronous glitch occurs while in synchronous, there is no problem of glitch. In asynchronous settling time is more while in synchronous settling time is less. Asynchronous counters are simple and straight in operation while synchronous are complex in operation.

Q.9. How many types of shift register counters are there and write their names also?

Ans. There are two types of shift register counters which are named as-

(a) Ring counter and
(b) Johnson counter.

Q.10. What is a register?

Ans. Register is a group of flip flop or binary cells which holds the binary information. Since a binary cell store a bit of information, n bit register has n flip flops and is capable of storing any information of n bit.